

Dataset Description for *Expansion on Exploring Common Trends in Online Educational Experiments*

This dataset is an expansion upon the dataset provided in [Exploring Common Trends in Online Educational Experiments Data](#). It contains the raw data for the problems logs and the students actions prior to the experiment period. The dataset itself is stored within a zip file called [expanded_experiment_dataset_2021-09-23.zip](#). The zip file contains experiment zip files named after the experiment they are represented for. These zips are also the same in the original dataset. The experiment zip files contains two CSVs files, each documented within its own section below.

exp_plogs_raw.csv

Within the ASSISTments platform, whenever a student starts a problem in the Student Workbench, a row in the problem logs database is added. This dataset represents the raw problem logs associated with each experiment, which can be associated with the `assistments\_reference\_problem\_log\_id` column in `exp\_plogs.csv` within the [original dataset](#).

id

The unique identifier that represents a problem a student has started.

assignment_log_id

The identifier of the assignment session that contained the problem the student is working on. If a teacher erases a student's progress on an assignment and completes the same problem, there will be two separate `assignment\_log\_id`s.

problem_id

The identifier of the problem the student was working on.

start_time

The time when the student started the problem. This is set when the problem has loaded. This is formatted in the [timestamp with time zone format](#).

end_time

The time when the student finished the problem. This exists when the student submits an answer for an ungraded response question, when they put in the correct answer, or when they exceed the maximum number of attempts for that problem. This is formatted in the [timestamp with time zone format](#).

continuous_score

The partial credit score of the student based upon the last answer submitted for this problem. The partial credit score is determined based upon the number of attempts and *student-supports* requested a student can make before completing the problem. Each time an incorrect answer is submitted or the student doesn't request the last *student-support* (e.g. an explanation or click the 'Show Answer' button after going through all the hints), the student's partial credit score will go down by a certain amount. This amount is determined by the number of attempts and *student-support* requests a student can make before receiving no credit (this is by default 3, it can be set to a different number by the teacher). If the student requests the last *student-support* (e.g. explanation or the answer after viewing all the hints), they will immediately receive no credit regardless of the number of attempts made or *student-supports* requested prior. This value will not exist if the answer has not been graded (e.g. open response that the teacher hasn't viewed).

discrete_score

An integer representation of the [continuous_score](#). This will be 0 if the partial credit score is not 1. This will not exist if the partial credit score does not exist.

answer_text

The last answer the student submitted for this problem. If the problem is an open response or a fill-in-the-blank question and the answer has been provided, then the answer text will be replaced with `

first_action_type_id

The identifier of the first action the student made on this problem. This will not exist if the student never performed an action. This can be one of the following values:

- 1: The student submitted an answer to the problem.
- 2: The student requested a hint from a hint set type *student-support* (through the 'Show Hint X of N' button) on the problem.
- 3: The student requested a scaffolding type *student-support* (through the 'Break this down into steps' button) on the problem.
- 4: The student requested the answer (through the 'Show Answer' button) to the problem.
- 5: The student requested an explanation type *student-support* (through the 'Show Explanation' button) on the problem.
- 6: The student clicked on the video (through the 'Show Video' button) on a problem.
- 7: The student requested support from an external live tutoring source (e.g. UpGrade) on a problem.

attempt_count

The number of attempts the student has made on this problem. If no attempts have been made, this value will be 0.

hint_count

The number of times a hint from a hint set type *student-support* has been requested by the student for a given problem. If no hints have been requested, this value will be 0.

bottom_hint

A true/false value which determines whether the student viewed the last *student-support* on this problem (e.g. clicked the 'Show Answer' or 'Show Explanation' button). If no *student-supports* have been requested, this value will be false.

first_response_time

The number of milliseconds from when the student started this problem to when the student made their first action on this problem (submitting an answer to this problem or requesting a *student-support*). If a student has never made an action on this problem, this value will not exist.

teacher_comment

The comment that the teacher has made on a student's work for this problem, typically used for open response questions. If a comment is present, the comment will be replaced with `` to avoid revealing any personal identifiers. If no comment has been made, this value will not exist.

prior_sactions.csv

Whenever a student does anything within the Student Workbench, an action is logged. An action can occur whenever a user starts an assignment, requests a *student-support*, submits an answer, etc. This dataset represents the actions of the students taken prior to the experiment period.

id

The unique identifier that represents an action the student had taken.

student_id

The identifier of the student that took the action.

timestamp

The time when the student performed the action. This is formatted in the [timestamp with time zone format](#).

assignment_log_id

The identifier of the assignment session that contained the action the student had performed. If a teacher erases a student's progress on an assignment and completes the same problem, there will be two separate `assignment\_log\_id`s.

problem_id

The identifier of the problem that contained the action the student performed. If the action was not performed on a problem-level (e.g. student starts an assignment, student starts a problem set), then this value will not exist.

used_as_scaffolding

A true/false value of whether the action performed was on a problem that was part of a scaffolding type *student-support*. If the action is not for a problem or on a problem that was part of a scaffolding type *student-support*, this will be false.

path

A string representing a URL of the identifiers for the context of this action (e.g. <problem set>/<problem>/<problem part>/<student-support>/<student-support type>).

tutor_uid_id

The identifier of the Student Workbench session the action was performed in. A session will always start when an assignment is started or resumed. If a student resumes the assignment at a later time or refreshes the browser, they will be assigned a new session identifier for the same assignment.

tutor_uid_sequence

The order in which an action is performed in a given [tutor_uid_id](#). Each session will start with a sequence of 1 and continue until the assignment is finished or the student starts a new session by resuming the assignment at a later time or through a browser refresh.

type

The type of action the student performed. This can be the following values:

- assignment_started
 - The student started the assignment. This will be the first action of a [tutor_uid_id](#) if the student has never opened the assignment before.

- `assignment_resumed`
 - The student resumed the assignment. This will be the first action of a [tutor uid id](#) if the student has opened the assignment before.
- `assignment_finished`
 - The student completed the assignment. This will be the last action in an [assignment log id](#). This may not be the last action in a [tutor uid id](#) if the student leaves the assignment or refreshes the browser, calling the ``assignment_resumed`` action.
- `problem_set_started`
 - The student started the problem set for an assignment or began a multi-part problem / scaffolding.
- `problem_set_resumed`
 - The student resumed the problem set for an assignment or came back to a multi-part problem / scaffolding.
- `problem_set_finished`
 - The student completed the problem set for an assignment or finished a multi-part problem / scaffolding.
- `problem_set_mastered`
 - The student completed three problems in a row (also known as a skill builder).
- `problem_set_exhausted`
 - The student has gone through all of the problems in a skillbuilder without completing three right in a row.
- `problem_limit_exceeded`
 - The student has exceeded the maximum number of problems they can complete for a skillbuilder in one day and must come back tomorrow to complete the assignment.
- `problem_started`
 - The student has started a problem in a problem set or a multi-part problem.
- `problem_resumed`
 - The student has resumed doing a problem in a problem set or a multi-part problem.
- `problem_finished`
 - The student has completed a problem in a problem set or a multi-part problem.
- `tutoring_set_started`
 - The student has requested a *student-support*.
- `tutoring_set_resumed`
 - The student has been given the *student-support* they requested after resuming the problem.
- `tutoring_set_finished`
 - The student has viewed all data within a *student-support* (e.g. all hints in a hint set).
- `hint_requested`
 - The student has requested a hint from a hint set type *student-support*.
- `scaffolding_requested`

- The student has requested a scaffolding type *student-support*. This will trigger the `problem\_started` or `problem\_set\_started` action with [used as scaffolding](#) returning true.
- `u_r_l_link_requested`
 - The student has requested a video type *student-support* through the 'Show Video' button.
- `explanation_requested`
 - The student has requested an explanation type *student-support*.
- `student_response`
 - The student has submitted an answer to the problem which can return the answer immediately.
- `student_submission`
 - The student has submitted an answer to the problem which must be graded by the teacher (open response).
- `answer_requested`
 - The student requested an answer type *student-support*.
- `user_selected_continue`
 - The student clicked the continue button to go to the next problem.
- `help_im_stuck`
 - The student completed a *student-support* (`tutoring\_set\_finished`) to see the answer and still does not get the answer correct. A stuck button appears in this case.
- `timer_started`
 - The student has begun a problem which has the timer active. This will occur after the `problem\_started` action.
- `timer_resumed`
 - The student has resumed a problem which has the timer active.
- `timer_paused`
 - The student has left the assignment while the timer was still active.
- `timer_finished`
 - The student has completed a problem which has the timer active. This will be triggered within a separate problem, so it will be after the next `problem\_started` action.
- `live_tutoring_requested`
 - The student requested external live tutoring through a third-party service (e.g. UpGrade).

`is_first_response`

A true/false value indicating whether the action performed was the first response made by the user (answer submitted or requested a *student-support*). This value does not exist if there is no [problem_id](#).

active_tutoring_type_id

An identifier which indicates what *student-support* can be requested for this problem by the student. This value will only be present if the [type](#) is `problem\_started`. Just because a value isn't listed does not mean you cannot get that type of *student-support* (e.g. you can get a hint set and a video at the same time). The values are as follows:

- 1: Hint set type *student-support*
- 2: Scaffolding type *student-support*
- 3: Video button ('Show Video')
- 5: Nothing
- 7: Explanation type *student-support*
- 8: Answer button ('Show Answer')
- 10: Feedback on what the student answers

is_pcs_eligible

A true/false value indicating if the student can receive a [continuous_score](#) for this problem. If not, then the value will be the same as the [discrete_score](#). This value will only be present if the [type](#) is `problem\_started`.

is_cb_viewable

A true/false value indicating if the student can see their [continuous_score](#) for this problem. If not, then the student will see the [discrete_score](#). This value will only be present if the [type](#) is `problem\_started`.

max_attempts_and_hints

An integer which represents the number of attempts and *student-supports* the student can request before they receive 0 credit on a problem. This value will only be present if the [type](#) is `problem\_started`.

show_correctness

A true/false value which indicates whether the student will know whether they got the answer correct on this problem. This value will only be present if the [type](#) is `problem\_started`.

scoring_ver

A string which represents the scoring algorithm used to determine the student's partial credit score. There is currently only one scoring algorithm: `v1`. This is calculated as $(\text{max_attempts_and_hints} - \text{current_attempt_hints} / \text{max_attempts_and_hints})$ where `current_attempt_hints` is the number of attempts or *student-supports* requested by the student. If the student requests the last *student-support* (explanation or 'Show Answer'), then this value will immediately become 0 regardless of the previous partial credit score. This value will only be present if the [type](#) is `problem\_started`.

hint_number

An integer indicating how many hints in a hint set type *student-support* the student has already viewed on this problem. This value will only be present if the [type](#) is ``hint_requested``.

total_hints

An integer indicating how many hints in a hint set type *student-support* the student can view for this problem. This is always the total number of hints in the *student-support*. This value will only be present if the [type](#) is ``hint_requested``.

response

The answer submitted by the student for this problem. If the problem is an open response, or if the problem is a fill-in-the-blank and the student got the answer wrong, the response will be ``<Blinded>`` to avoid revealing any personal identifiers. This value will only be present if the [type](#) is ``student_response`` or ``student_submission``.

correct

A true/false value indicating whether the student got the answer correct for this problem, assuming it was a problem that could be graded immediately. This value will only be present if the [type](#) is ``student_response``.

continuous_score

Same as [continuous_score](#) in the other dataset. This value is reflective based upon where in the problem they are (e.g. how many attempts made, *student-supports* requested). This value is only present if the [type](#) is ``problem_finished``, ``hint_requested``, or ``student_response``.

discrete_score

Same as [discrete_score](#) in the other dataset. This value is reflective based upon where in the problem they are (e.g. how many attempts made, *student-supports* requested). This value is only present if the [type](#) is ``problem_finished``, ``hint_requested``, or ``student_response``.

icon

A value of what icon the student will be displayed after completing the problem. This value is only present if the [type](#) is ``problem_finished``. The values are as follows:

- GREEN_CHECK
 - The student got the answer right with full credit.
- RED_CROSS
 - The student received no credit for the problem.
- HIGHLIGHTED_RED_CROSS
 - The student received no credit for the problem after requesting a *student-support*.
- BLUE_DOT

- The student was not shown the answer to the problem (open response or [show correctness](#) is false for the problem).
- GREEN_CROSS
 - The student got the answer right but did not receive full credit.